



Module 1: Cyber Security Fundamentals

Module overview: This module introduces the core concepts, principles, and operating models of cyber security so learners can understand why security matters, how it supports the business, and what good security practice looks like in a modern organisation.

Learning objectives:

- Explain cyber security in business terms.
- Describe the CIA Triad and the People, Process, and Technology model.
- Identify the purpose of key cyber security domains.
- Recognise why training and awareness are critical to organisational resilience.



Cyber Security has become one of the most critical business disciplines of the modern era. Once regarded primarily as a technical function managed by IT departments, it has evolved into a strategic organisational capability that directly impacts operational resilience, customer trust, regulatory compliance, financial stability, and long-term business success.

As organisations continue to digitise their operations, adopt cloud technologies, support distributed workforces, integrate with third-party ecosystems, and embrace emerging technologies such as artificial intelligence and automation, cyber security has become deeply embedded in almost every aspect of the modern enterprise.

Today, effective cyber security is not only about protecting systems and data — it is about enabling secure business operations, maintaining stakeholder confidence, supporting innovation, and ensuring organisations can operate resiliently in an increasingly connected and constantly evolving digital world.

Learner summary: Cyber security is no longer only a technical concern; it is a core business capability that supports resilience, trust, compliance, and long-term performance.

1. The Security Operating Mechanism

The Security Operating Mechanism



At the core of every effective cyber security strategy lies a simple yet powerful operating mechanism — balancing the protection of information, the enablement of business operations, and the management of human behaviour. Although cyber security is often associated with technical tools and complex systems, true organisational resilience is achieved through a combination of foundational security principles, operational discipline, and strategic alignment.

The Security Operating Mechanism provides a practical framework for understanding how cyber security functions within a modern organisation. At its foundation is the CIA Triad, which defines the three core objectives of cyber security: Confidentiality, Integrity, and Availability. These principles guide how organisations protect information, maintain trust in systems and data, and ensure critical services remain accessible when needed.

Supporting the CIA Triad is the People, Process, and Technology (PPT) model, which represents the operational pillars required to implement, manage, and sustain cyber security effectively in the real world. Together, these elements create a balanced approach that combines governance, human awareness, operational processes, and technology-driven controls.

Collectively, these models help organisations move beyond reactive security practices toward a more strategic, resilient, and business-aligned security posture.

Learner summary: Effective cyber security depends on balancing foundational principles with people, processes, and technology to support business-aligned resilience.

2. The 12 Cyber Security Domains



Cyber Security is a complex and constantly evolving discipline that can be viewed from many different perspectives. From a business standpoint, it involves understanding the threat landscape, identifying organisational risks, and implementing the appropriate measures, including governance, policies, processes, and technologies, to reduce exposure and strengthen resilience.

To simplify this complexity, cyber security is often divided into a number of specialised domains, each playing a critical role in creating and maintaining a secure environment.

These domains are not fixed or universally defined. Different industry frameworks, standards, and organisations may structure or categorise them differently. However, the following 12 domains provide a practical view of the key components of a modern cyber security programme:

- **Governance, Risk & Compliance (GRC)**: Establishes policies, oversight, legal alignment, and risk management across the organisation.
- **Identity & Access Security**: Controls who can access systems, applications, and data, and under what conditions.
- **Network Security**: Protects network infrastructure, traffic, and connectivity from unauthorised access and attack.
- **Endpoint Security**: Secures laptops, desktops, mobile devices, and other user endpoints.
- **Cloud Security**: Protects cloud platforms, services, workloads, and configurations.
- **Application Security**: Reduces vulnerabilities in software, systems development, and deployed applications.
- **Data Protection**: Safeguards sensitive information through classification, encryption, handling, and loss prevention controls.

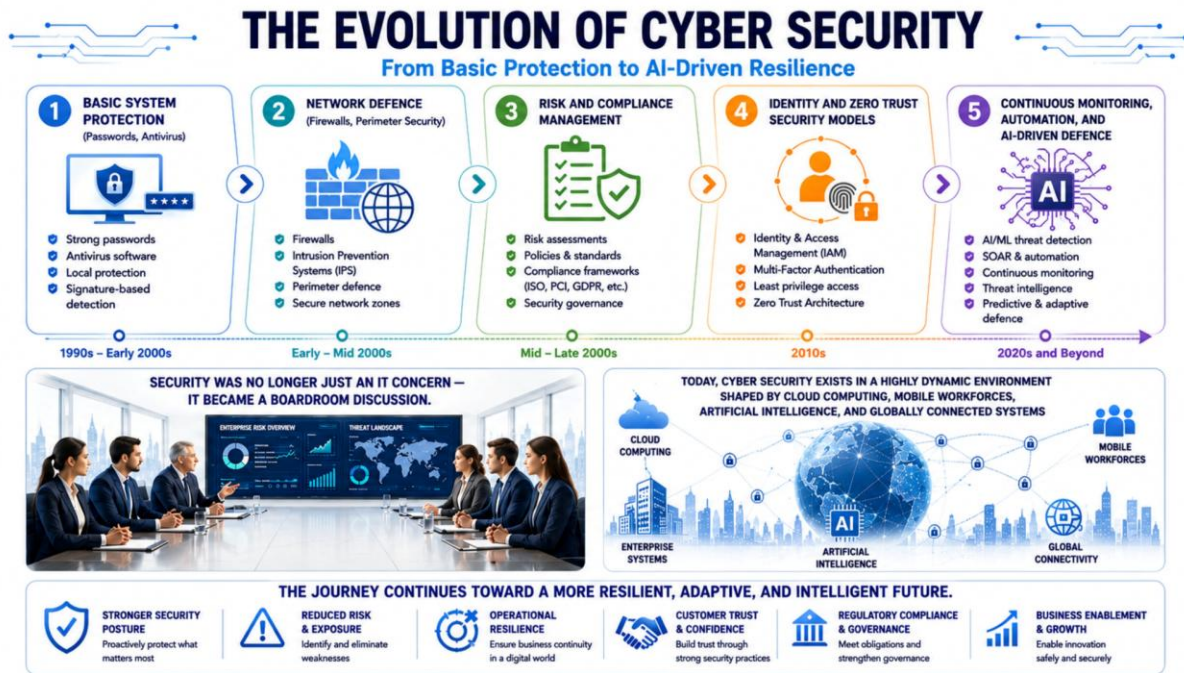
- **Security Operations:** Monitors, detects, investigates, and responds to threats and incidents.
- **Threat & Vulnerability Management:** Identifies weaknesses, emerging threats, and remediation priorities before they are exploited.
- **Third-Party & Managed Services Security:** Manages risk introduced by vendors, partners, and outsourced technology services.
- **Business Continuity & Incident Response:** Prepares the organisation to contain incidents, recover operations, and maintain resilience.
- **Security Awareness & Culture:** Builds secure behaviour, accountability, and shared responsibility across the workforce.

While each domain serves a distinct purpose, they are highly interconnected, and together they form the operational, strategic, and cultural foundation required to achieve modern cyber resilience.

Cyber Security is no longer focused solely on protecting systems and infrastructure. It is about protecting business value, enabling trusted digital transformation, supporting operational continuity, and building organisational resilience in an increasingly complex threat landscape.

Learner summary: Cyber security domains break a complex discipline into manageable areas of focus that together create a complete security programme.

3. The Evolution of Cyber Security



Cyber Security has evolved dramatically over the past few decades, transforming from a largely technical function into a critical business discipline. In its early stages, security focused primarily on basic protective measures such as passwords, user authentication, and isolated system controls. As networks expanded and the internet became widely adopted during the 1990s, organisations faced a growing range of threats, including viruses, worms, malware, and early forms of cyber intrusion. This period marked the emergence of firewalls and antivirus technologies as the first generation of cyber defence mechanisms.

As digital transformation accelerated in the early 2000s, organisations began to recognise that cyber incidents could result in significant financial loss, operational disruption, reputational damage, and regulatory exposure. In response, structured security frameworks, governance models, and compliance requirements such as ISO 27001, GDPR, and POPIA became increasingly important. Cyber Security was no longer viewed solely as an IT responsibility; it became a strategic business concern discussed at executive and board level.

Today, cyber security operates within a highly interconnected and rapidly evolving digital landscape shaped by cloud computing, mobile and remote workforces, artificial intelligence, automation, and globally connected systems. Modern threat actors are organised, sophisticated, well-funded, and continuously adapting their tactics. As a result, cyber security is no longer focused on static protection alone, but on continuous monitoring, resilience, adaptation, and the ability to respond effectively to an ever-changing threat environment.

In simple terms, cyber security has evolved through:

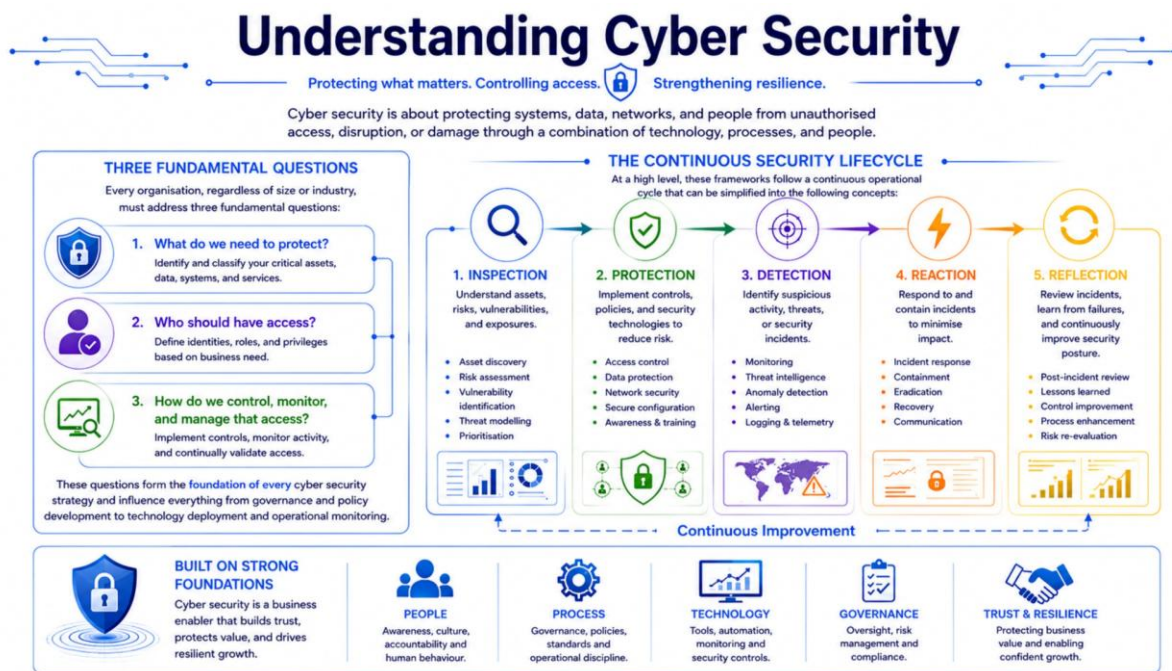
- Basic system protection (passwords, antivirus)
- Network defence (firewalls, perimeter security)
- Risk and compliance management
- Identity and Zero Trust security models

- Continuous monitoring, automation, and AI-driven defence

In reality, Cyber Security is an ever-changing landscape where both technology and threats evolve daily.

Learner summary: Cyber security has evolved from basic technical protection into a continuous, strategic discipline shaped by business risk and digital change.

4. Understanding Cyber Security



At its core, Cyber Security is about protecting systems, networks, applications, data, and people from unauthorised access, disruption, misuse, or damage. However, modern cyber security extends far beyond technology alone. It represents a combination of people, processes, governance, and technology working together to protect organisational operations, maintain trust, and support business resilience in an increasingly connected world.

Every organisation, regardless of size or industry, must address three fundamental questions:

- What do we need to protect?
- Who should have access?
- How do we control, monitor, and manage that access?

These questions form the foundation of every cyber security strategy and influence everything from governance and policy development to technology deployment and operational monitoring.

Understanding what needs to be protected requires organisations to identify their critical assets. These may include sensitive data, intellectual property, customer information, financial systems, operational technology, cloud environments, or business-critical services. Once these assets are identified, organisations must determine who requires access to them and under what conditions. This introduces the need for identity management, access control, authentication, and ongoing monitoring to ensure access remains appropriate, secure, and aligned with business requirements.

Modern cyber security also relies heavily on structured security frameworks that help organisations manage risk and improve resilience in a consistent and measurable way. One of the most widely recognised examples is the National Institute of Standards and Technology (NIST) Cybersecurity Framework, which provides a practical approach for managing cyber security across the organisation.

At a high level, these frameworks often follow a continuous operational cycle that can be simplified into the following concepts:

Inspection – understanding assets, risks, vulnerabilities, and exposures

Protection – implementing controls, policies, and security technologies to reduce risk

Detection – identifying suspicious activity, threats, or security incidents

Reaction – responding to and containing incidents to minimise impact

Reflection – reviewing incidents, learning from failures, and continuously improving security posture

This highlights an important reality of modern cyber security: security is not a once-off project or a static technical control. It is a continuous process of assessment, protection, monitoring, response, and improvement.

As organisations continue to evolve digitally, cyber security must evolve alongside them. Effective cyber security therefore requires strategic thinking, operational discipline, user awareness, executive support, and continuous adaptation to an ever-changing threat landscape.

Learner summary: Modern cyber security is a continuous cycle of identifying what matters, controlling access, managing risk, and improving over time.

5. The CIA Triad (Confidentiality, Integrity, Availability)



The CIA Triad represents the three foundational principles of modern cyber security. These principles guide how organisations protect information, maintain trust in their systems, and ensure business operations remain secure and reliable.

Confidentiality ensures that sensitive information is only accessible to authorised individuals.

Integrity ensures that data remains accurate, trustworthy, and protected from unauthorised modification.

Availability ensures that systems, applications, and information remain accessible when needed by the business.

Together, these three principles form the foundation upon which cyber security strategies, policies, controls, and technologies are built.

Simple View:

Confidentiality = "Who can see it?"

Integrity = "Can it be trusted?"

Availability = "Can I access it?"

Learner summary: The CIA Triad provides the core lens for understanding how security protects information, trust, and business continuity.



Effective cyber security depends on achieving the right balance between People, Process, and Technology. These three pillars work together to create a secure, resilient, and well-governed environment.

People represent the human element of cyber security, including user behaviour, awareness, accountability, and organisational culture.

Process includes the governance structures, policies, standards, procedures, and operational practices that guide how security is managed and enforced.

Technology refers to the tools, platforms, automation, and monitoring capabilities used to protect systems, detect threats, and enforce security controls.

While technology plays a critical role in modern cyber defence, many security incidents and breaches still originate from the People layer through human error, weak passwords, phishing attacks, social engineering, or poor security awareness. This highlights the importance of building a strong security culture alongside effective processes and technologies.

Together, People, Process, and Technology form the operational foundation required to support a mature and effective cyber security strategy.

Learner summary: Strong cyber security depends on more than tools; people, processes, and technology must work together to produce consistent security outcomes.



Training and awareness are among the most critical, and often most underestimated components of modern cyber security. While technology can detect and block many threats, it is ultimately people who make decisions, click links, share information, approve transactions, and grant access to systems and data.

For this reason, training and awareness represent the first line of defence in cyber security and are often referred to as the Human Firewall.

Within the People, Process, and Technology (PPT) model, training and awareness form the practical implementation of the “People” pillar. They also directly support the CIA Triad, particularly Confidentiality and Integrity, by helping users understand how to protect sensitive information and maintain trust in organisational systems and data.

Effective security awareness programmes typically include:

- Security awareness training covering topics such as phishing, password security, and secure data handling
- Simulated phishing campaigns to test and strengthen user awareness
- Role-based training tailored for executives, developers, IT teams, and other business functions
- Continuous reinforcement through micro-learning, reminders, campaigns, and ongoing communication
- Measurable improvement through metrics such as click rates, reporting rates, and behavioural change

Why It Matters

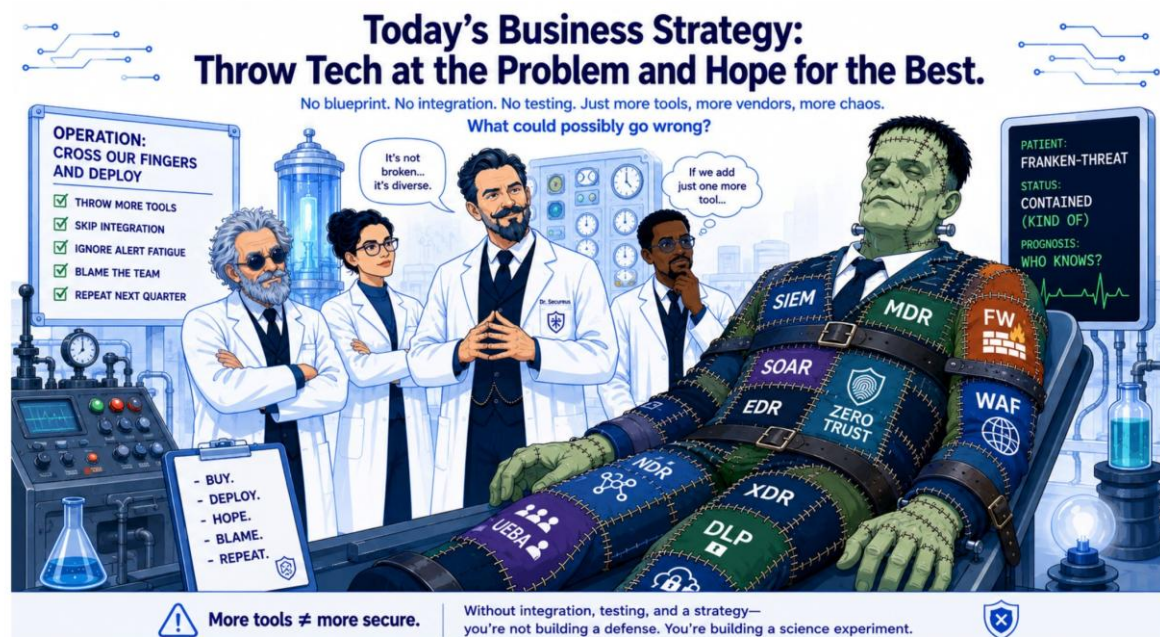
- Most cyber-attacks begin with some form of human interaction, particularly phishing and social engineering
- Even the most advanced technologies can be bypassed if users are unaware or unprepared
- Effective awareness programmes transform employees from potential security risks into active participants in organisational defence

A strong security culture does not happen by accident. It is developed through continuous education, reinforcement, accountability, and leadership support.

Your employees are either your biggest vulnerability or your strongest defence and training determines which.

Learner summary: Security awareness is essential because human behaviour can either strengthen or weaken every other control in the environment.

In this module, you learned that cyber security is a business-critical discipline built on foundational principles, supported by people, processes, and technology, and delivered through interconnected security domains that strengthen organisational resilience.



Summative assessment:

1. Why is cyber security now considered a business discipline rather than only an IT responsibility?
Cyber security is a business discipline because cyber risk affects operations, compliance, finances, reputation, and customer trust.
2. How do the CIA Triad and the PPT model work together to strengthen an organisation's security posture?
The CIA Triad defines what must be protected, while PPT explains how people, processes, and technology work together to protect it.
3. Why is it important to understand cyber security as a set of interconnected domains rather than isolated technical controls?
Domains are interconnected because effective security depends on coordinated controls rather than isolated technical measures.
4. What changes in technology and business operations have made cyber security more strategic over time?
Cyber security became more strategic due to digital transformation, cloud adoption, remote work, regulation, and increasingly sophisticated threats.
5. What are the three fundamental questions every organisation must answer when building a cyber security strategy?

The three questions are: what do we need to protect, who should have access, and how do we control and monitor that access.

6. How would you explain confidentiality, integrity, and availability in simple business language?

Confidentiality means only the right people can see information, integrity means the information can be trusted, and availability means it can be accessed when needed.

7. Why can strong technology controls still fail when people and processes are weak?

Technology can fail when people make poor decisions or when processes are weak, inconsistent, or not enforced.

8. What behaviours should a strong security awareness programme change in everyday employee activity?

A strong awareness programme should improve behaviours such as recognising phishing, protecting passwords, handling data carefully, and reporting suspicious activity quickly.